larger, as shown in Fig. 25-9b. However, the signal is becoming larger *faster* than the noise, resulting in an overall improvement in the SNR. Don't be confused into thinking that a lower signal will provide less noise and therefore better information. Remember, your goal is *not* to reduce the noise, but to extract a signal *from* the noise. This makes the SNR the key parameter.

Many imaging systems operate by converting one particle type to another. For example, consider what happens in a medical x-ray imaging system. Within an x-ray tube, *electrons* strike a metal target, producing *x-rays*. After passing through the patient, the x-rays strike a vacuum tube detector known as an image intensifier. Here the x-rays are subsequently converted into *light photons*, then *electrons*, and then back to *light photons*. These light photons enter the camera where they are converted into *electrons* in the well of a CCD. In each of these intermediate forms, the image is represented by a finite number of particles, resulting in added noise as dictated by Eq. 25-1. The final SNR reflects the combined noise of *all* stages; however, one stage is usually dominant. This is the stage with the *worst* SNR because it has the *fewest* particles. This limiting stage is called the **quantum sink**.

In night vision systems, the quantum sink is the number of light photons that can be captured by the camera. The darker the night, the noisier the final image. Medical x-ray imaging is a similar example; the quantum sink is the number of x-rays striking the detector. Higher radiation levels provide less noisy images at the expense of more radiation to the patient.

When is the noise from Poisson statistics the primary noise in an image? It is dominant whenever the noise resulting from the quantum sink is greater than the other sources of noise in the system, such as from the electronics. For example, consider a typical CCD camera with an SNR of 300. That is, the noise from the CCD preamplifier is 1/300th of the full scale signal. An equivalent noise would be produced if the quantum sink of the system contains 90,000 particles per pixel. If the quantum sink has a smaller number of particles, Poisson noise will dominate the system. If the quantum sink has a larger number of particles, the preamplifier noise will be predominant. Accordingly, most CCD's are designed with a full well capacity of 100,000 to 1,000,000 electrons, minimizing the Poisson noise.

Morphological Image Processing

The identification of objects within an image can be a very difficult task. One way to simplify the problem is to change the grayscale image into a **binary image**, in which each pixel is restricted to a value of either 0 or 1. The techniques used on these binary images go by such names as: **blob analysis**, **connectivity analysis**, and **morphological image processing** (from the Greek word *morphe*, meaning shape or form). The foundation of morphological processing is in the mathematically rigorous field of *set theory*; however, this level of sophistication is seldom needed. Most morphological algorithms are simple logic operations and very *ad hoc*. In